

Optimal Head-mounted IMU Placement for Heart Rate Detection Using Ballistography

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Abstract—Continuous and accurate heart rate monitoring is pivotal for the early detection of cardiovascular conditions. Traditional heart rate sensing methods, such as the use of adhesive electrocardiogram (ECG) electrodes, often fall short in terms of comfort and convenience for everyday wear and continuous monitoring. Ballistocardiography (BCG), a non-invasive alternative, offers a promising solution by measuring the body's mechanical reactions to the ejected blood by the heart during cardiac cycles. This paper introduces a novel heart detection method utilizing IMU-based BCG applied to the head, focusing on optimizing sensor placement on the head for improved accuracy. Noteworthy is that the detection algorithm needs to be robust against noise and movement artifacts; therefore, the choice of sensor location and signal processing techniques become key factors. This pilot study explores various IMU mounting locations on the head and signal processing techniques to improve the feasibility and accuracy of head-mounted wearable devices employing BCG.

Keywords—Ballistocardiography, Inertial Measurement Unit, Kalman Filter, Heart Rate, Data Fusion

I. INTRODUCTION

Cardiac arrests and heart diseases are the leading cause of death in the United States (1 in 5 deaths in 2021), and one fifth of the heart attacks are silent, which means that the person is not aware of the damage that has been done to their body [1]. Hence, early detection and diagnosis of cardiovascular diseases and cardiac arrhythmia is crucial for the public health [2]. One solution is to continuously monitor heart rate and heart condition, especially in people with high risk of heart attacks. Cardiac autonomic function (CAF), one of the cardiac activities, is regulated by autonomic nervous system (ANS) and controls the heart functions. One of the non-invasive methods for cardiac activity analysis is monitoring heart rate (HR) and heart rate variabilities (HRV), which are reliable indicators for CAF and ANS assessments [3] and are introduced as important predictors for heart failure detection in previous studies [4-6]. The traditional method for collecting heart rate data is to use electrocardiogram (ECG) adhesive electrodes on the chest, which fall short in terms of comfort and convenience for everyday wear and continuous monitoring. An alternative method for ECG is using ballistocardiography (BCG). BCG is a non-invasive technique that measures repeated mechanical reactions of the body due to the ejected blood by the heart in

cardiac cycles [7]. According to recent studies, inertial measurement unit (IMU)-based BCG sensors can be used to estimate HR and HRV [8]. Traditional BCG devices are mostly embedded into the person's bed or mattress to monitor heart rate during sleep. Other recent studies, however, have shown acceptable results using BCG wearables with embedded IMUs to detect HR in a connected and continuous way [9]. Each wearable device proposed in the literature covers different yet valuable physiological monitoring aspects. However, head-mounted wearables offer unique advantages. Their proximity to the ear and to the carotid and temporal arteries for acquiring heartbeats with reduced motion artifacts, making them suitable for potential long-term, non-invasive, and unobtrusive monitoring [10]. For example, Crum refers to the ear as the biological equivalent of a USB port, from which multiple physiological signals and vital signs can be derived [11]. This provides an added advantage in terms of wearability, given its considerable synergies with hearing aids, earbuds, and headphones that consumers use daily. Therefore, in this pilot study, we introduced a heart rate detection algorithm using a head-mounted IMU sensor at different locations, which could eventually be integrated into earbuds and headphones products to continuously measure HRV.

BCG is a well-known non-invasive method to capture the repeated micromovements of the body due to the cardiovascular activity [12]. The important wave that should be detected in BCG signal for calculating HR is J peak. J peak is the largest head-ward wave that shows the aortic valve opening and is a delayed wave form of R peak, the critical peak that is usually used for HR detection, in ECG signals. Therefore, by estimating J peak in the BCG, one could find HR and HRV in the signals [10,11]. Different methods were developed in literature to estimate the HR signals from accelerometers and gyroscopes in IMU sensors. For example, DeMello et al. argued in their BCG-based HR study that combining seismocardiography and gyrocardiography signals improves the accuracy of HR monitoring [13]. Da He et al. developed an ear worn BCG cardiovascular monitoring method using accelerometer data and compared the head BCG results with head ECG results [2,8]. Another IMU-based BCG study used both accelerometer and gyroscope signals and introduced heart and respiratory rate estimation using moving windows [14]. The authors developed their BCG system using the z-axis signal from accelerometer data and y-axis signal from gyroscope data; where the z-axis

corresponded to the dorso–ventral axis (back-to-front) of the body, which is perpendicular to the sagittal plane of the body, and y-axis corresponded to the superior–inferior axis (head-to-foot), which is perpendicular to the vertical plane of the body. Henandez et al. presented HR estimation using IMU sensors in Google Glass [9]. Milena et al. extracted the linear and nonlinear HRV indices from accelerometer and gyroscope signals extracted from chest-mounted sensors, and concluded that using signals from both accelerometer and gyroscope results in more accurate outcomes in lying positions [12]. Gardner et al. showed that HR can be detected using signals from a gyroscope embedded in a CPAP (Continuous Positive Airway Pressure) mask [17,18]. Although a few studies have already introduced HR detection using the signals collected from chest and head regions [9], no study has yet investigated the optimal location on the head for robust HR monitoring using IMU-based BCG.

Robust HR estimation is crucial in health monitoring especially when the signals are extracted from wearable devices. The collected data from wearable devices are prone to have low signal-to-noise ratio due to noise and movement artifacts. Different strategies have been introduced in the literature to overcome these issues and to acquire more robust signals for HR detection. These methods include averaging, signal fusion, machine learning, signal quality assessment, among others [19]. Machine learning methods can detect and discard the artifacts in an effective way, but they need to have a large amount of training data for accurate results. In contrast, data fusion techniques including Kalman filter (KF), custom weighted average, a trimmed mean filter, and Bayesian fusion have shown promising results to detect patterns in the signals and removing abrupt movement artifacts even, with small dataset [19]. The KF-based data fusion technique is introduced as the most effective among these approaches [20].

In this study, we applied a KF to estimate HR using gyroscope and accelerometer signals extracted from an IMU sensor placed at various head locations. The rest of the paper is organized as follows. Section II explains the dataset and experimental methodology and Section III presents the results. Section IV discusses the results of the proposed algorithm and the comparison of different locations, and Section V concludes the paper.

II. METHODS

A. Dataset

The gyroscope and accelerometer signals were extracted from a 31-year-old female subject (height: 156 cm, weight: 52 kg) with no history of heart condition, in a seated position using a single Shimmer3 IMU (Shimmer, Dublin, Ireland) by placing the sensor at four locations on the head in sequence. These four locations included: left temple close to the eye, in front of the left ear on the zygomatic bone, behind the left ear on the temporal bone (mastoid), and on the left side of the neck on the left common carotid artery as illustrated in Fig. 1. These locations were chosen due to their proximity to pulse detection sites on the body. Pulse is the expansion and contraction of the arterial wall caused by the repeated contractions of the heart and is readily recognizable on locations which are closer to the arteries. These locations consist of radial artery on the wrist,

carotid artery on the neck, temporal artery on the side of the eye, facial artery behind the ear, etc. [21]. Therefore, the mentioned four locations were chosen based on their proximity to the main arteries of the body which can better show the vibrations induced by the heart. The subject performed the experiment 3 times (each 4 minutes long) for each sensor location. The study protocol was approved by the Research Ethics Board of Simon Fraser University (ethics number: 30001736).

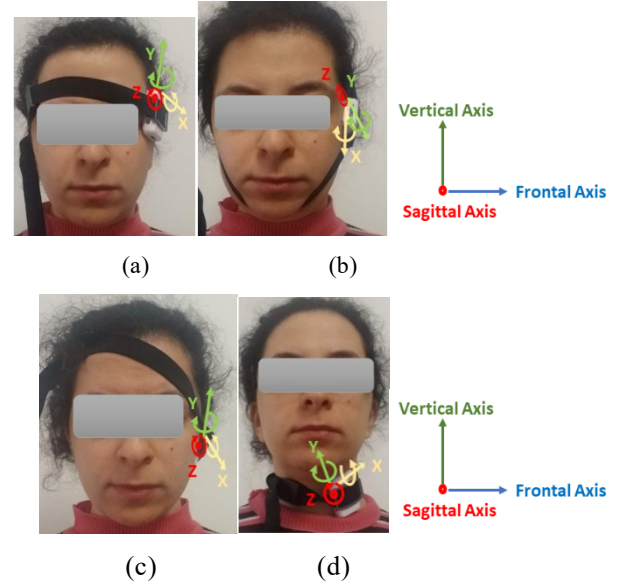


Figure 1. The four sensor placement locations on the head with respect to the body axes shown on the right side: (a) left temple close to the eye, (b) in front of the left ear on the zygomatic bone, (c) behind the left ear on the temporal bone (mastoid), and (d) on the left side of the neck on the left common carotid artery.

The wearable Shimmer3 IMU is equipped with a tri-axial accelerometer (full scale $\pm 16g$) and a tri-axial gyroscope (full scale 2000 degree/sec) [22]. The sensor was attached to the different locations on the head using a strap, as shown in Fig. 1. A sampling frequency of 512 Hz was chosen, consistent with that of the ECG reference sensor (Shimmer3 ECG sensor). Different signal processing and filtering methods were then used to detect HR in each axis of the accelerometer and gyroscope, as well as their normalized signals that made up the fourth signal. First, different signal processing algorithms were applied to all signals to detect HR (J peaks). Next, the detected HR signals were used for estimating the heartbeats using the KF, to determine which accelerometer or gyroscope signals perform the best in each location.

B. Heart Rate Detection Algorithm

The HR detection algorithm, which is fully described in [17], consists of multiple signal processing steps. This process is applied to both gyroscope and accelerometer signals of the IMU. First, a normalized signal was calculated using

$$g_n = \sqrt{g_x^2 + g_y^2 + g_z^2} \quad (1)$$

where g_x , g_y , g_z , and g_n are x, y, z, and normalized components of IMU signals, respectively. Next, Hilbert transformation was applied to all signals except the normalized

signal. After that, a 2nd order Butterworth filter with cut off frequencies of 0.5 and 10 Hz is used to remove motion artifacts in all signals including the normalized signal. In addition, a threshold is set to remove large movements for each signal. This threshold is calculated by defining 6 non-overlapping consecutive windows (each window has width of 1.5 seconds). In each window, the maximum value is calculated, and the average value of all these four maximum values is defined as the threshold for that axis. Finally, the J peaks in all four signals are detected using Pan-Tompkin method [23]. Furthermore, the Pan-Tompkin method is also used to determine the R peaks in the ECG reference signals.

C. Kalman Filter-based Data Fusion

KF is used here to estimate heartbeats in a robust and accurate way as the signals are from multiple sources based on a similar study [17]. The detected HRs from previous section are the inputs to the KF, which are implemented twice since there are two HR datasets, one from the gyroscope and the other from the accelerometer. However, the procedure is the same for both. The period is divided into non-overlapping 1.5 seconds windows. Each 1.5 seconds window needs to have at least one HR because a normal person has HR of 40 BPM (beats per minute) or higher. Thresholds of 40 BPM and 150 BPM were defined to remove the outliers from the dataset. The KF consists of two stages. The first stage is the prediction step, which is based on a defined model of the state. The other stage is the update step, which is based upon the observation measurements. We assume HR will remain constant, therefore for time k , the HR is

$$HR_k = HR_{k-1} + w_k \quad (2)$$

where w_k is the variation of the HR, which is assumed to be a Gaussian noise with zero mean and covariance Q_k .

The observation measurement is defined as

$$z_k = H_k x_k + v_k \quad (3)$$

where v_k is the observation noise which is defined as a zero mean Gaussian noise with noise of R_k . H_k and x_k are defined as follows

$$H_k = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad (4)$$

$$x_k = \begin{bmatrix} g_x \\ g_y \\ g_z \\ g_n \end{bmatrix} \quad (5)$$

where g_x, g_y, g_z, g_n are HR estimations from the x, y, z, normalized components, respectively.

The KF's prediction equations are then defined as

$$\widehat{x_{k|k-1}} = \widehat{x_{k-1|k-1}} \quad (6)$$

$$P_{k|k-1} = P_{k-1|k-1} + Q_k \quad (7)$$

where $\widehat{x_{k|k-1}}$ is the estimation of HR at time k based on the observation known at time $k-1$. P is the measure of accuracy of the prediction. The update equations are as follows

$$y_k = z_k - H_k \widehat{x_{k|k-1}} \quad (8)$$

$$S_k = H_k P_{k|k-1} H_k^T + R_k \quad (9)$$

$$K_k = P_{k|k-1} H_k^T S_k^{-1} \quad (10)$$

$$\widehat{x_{k|k}} = \widehat{x_{k|k-1}} + K_k y_k \quad (11)$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \quad (12)$$

The initial estimation for HR is defined by the ECG reference to obtain more accurate results. The Q_k is set to 0.3 based on similar studies [17,18], and the initial value for P is set to 0.1. R_k is defined as

$$\begin{bmatrix} \sigma_{x,k} & 0 & 0 & 0 \\ 0 & \sigma_{y,k} & 0 & 0 \\ 0 & 0 & \sigma_{z,k} & 0 \\ 0 & 0 & 0 & \sigma_{n,k} \end{bmatrix} \quad (13)$$

where

$$\sigma_k = |g_k - \widehat{x_{k-1|k-1}}| \quad (14)$$

The initial σ_k is set to 0.01. Moreover, if no HR is detected in a window, σ_k is set to 0.01 again to avoid issues with inverting a matrix with zero values (i.e., the S matrix).

Finally, the KF is applied to both accelerometer and gyroscope-extracted HR as mentioned, and root mean square error (RMSE) is calculated to compare the results:

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (x_i - \widehat{x}_i)^2}{N}} \quad (15)$$

where x_i is the true value and $\widehat{x}_i$ is the estimated value. The estimation error is calculated for both estimation results, which are derived from the accelerometer and gyroscope signals of the IMU. These estimation errors are used to compare the results of different mounting locations on the head, as well as the type of signal that was used for the estimation process.

III. RESULTS

Figure 2 illustrates a sample result using the proposed HR detection algorithm on the gyroscope's raw z-axis signal. This result shows that by filtering unwanted frequencies, using Hilbert transformation method, and applying thresholds, a noisy gyroscope signal from the IMU is transformed to a signal from which J peaks are easier to detect. However, there are no large movements in this sample result, and the signal remains unchanged after implementing the threshold. Figure 3 shows the accelerometer's KF estimation result, which is compared against the four accelerometer axes (x, y, z, and normalized) and the ECG reference signal. This figure shows the importance of using multiple signals instead of one single signal for HR detection. The HRs detected from the accelerometer signals, which consist of the accelerometer's x-axis (HR-BCG-AccX), y-axis (HR-BCG-AccY), z-axis (HR-BCG-AccZ), and the normalized signal of all accelerometer axes (HR-BCG-AccN) are not very accurate individually; however, when the KF is applied, the

estimation HR (Estimated-HR-BCG) is closer to the ECG reference (HR-ECG).

Table I shows the average RMSE of the three HR estimation experiments conducted at each of the four locations of the head. In this table, RMSE- ECG (Acc), RMSE- Ave (Acc), RMSE- ECG (Gyro), and RMSE- Ave (Gyro) are defined as (i) the RMS error between the accelerometer-based KF estimation and the ECG reference, (ii) the RMS error between the accelerometer-based KF estimation and the average of all accelerometer signals, (iii) the gyroscope-based estimation and the ECG reference, and (iv) the RMS error between the gyroscope KF estimation and the average of all gyroscope signals, respectively. Based on this table, the HR estimation when the IMU is attached to the neck shows the least error when using the accelerometer signals.

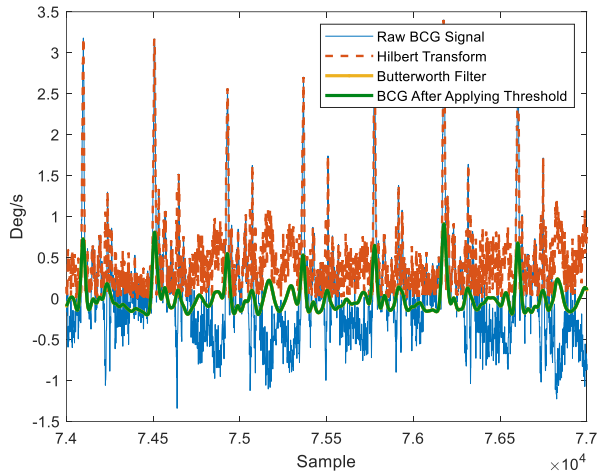


Figure 2. The gyroscope's raw z-axis signal and its filtered signals.

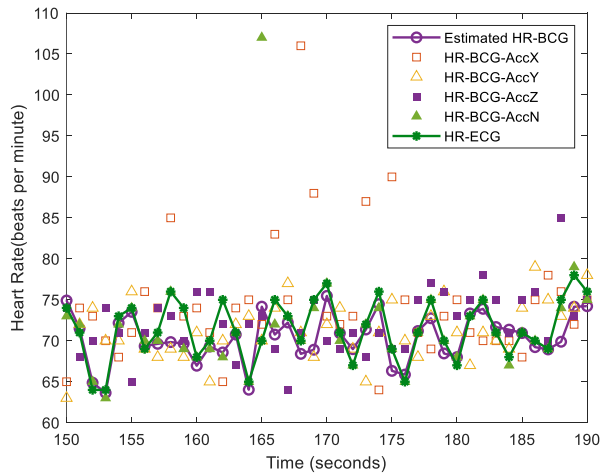


Figure 3. The estimated HR signal using the accelerometer data of the IMU compared to the ECG reference signal.

Table 1. RMSE comparison between four head mounting locations.

	Neck on Carotid	Back on Mastoid	Front on Zygomatic	Temple
RMSE-ECG (Acc) (BPM)	8.3980 ± 3.3837	21.1240 ± 1.3576	22.2865 ± 1.8312	20.4969 ± 0.4263
RMSE-Ave (Acc) (BPM)	4.2350 ± 0.4779	36.7901 ± 0.7751	23.5696 ± 5.1005	30.6552 ± 3.0526
RMSE-ECG (Gyro) (BPM)	18.6711 ± 0.2252	21.1801 ± 6.1974	16.6713 ± 1.3691	16.3022 ± 2.8810
RMSE-Ave (Gyro) (BPM)	24.7917 ± 15.2274	26.6923 ± 17.1239	10.4351 ± 4.5242	31.8914 ± 14.2556

Table 2 shows further analysis on the RMSE values with respect to each accelerometer axis at the neck location. In this table, RMSE (x axis), RMSE (y axis), RMSE (z axis), RMSE (normalized signal), RMSE (ECG signal), and RMSE (Ave of axes) are defined as (i) the RMS error between the accelerometer-based KF estimation and its x-axis signal, (ii) the RMS error between the accelerometer's KF estimation and its y-axis signal, (iii) the RMS error between the accelerometer's KF estimation and its z-axis signal, (iv) the RMS error between the accelerometer's KF estimation and its normalized signal, (v) the RMS error between the accelerometer's KF estimation and (vi) the ECG reference, and the RMS error between the accelerometer's KF estimation and the average of all accelerometer signals, respectively. As it can be seen in this table, the accelerometer's z-axis signal shows the best result. The direction of this axis with respect to the body is shown in Fig. 1(d).

Table 2. RMSE comparison between different axes at neck location

	1 st Trial	2 nd Trial	3 rd Trial	Mean of RMSE
RMSE (x axis) (BPM)	24.9825	10.9203	17.8003	17.9010 ± 7.0316
RMSE (y axis) (BPM)	4.7268	6.1937	17.3770	9.4325 ± 6.9191
RMSE (z axis) (BPM)	6.0496	5.7505	8.8139	6.8713± 1.6889
RMSE (normalized signal) (BPM)	21.4443	3.7032	3.1541	9.4339 ± 10.4050
RMSE (ECG signal) (BPM)	11.8689	5.1088	8.2163	8.3980 ± 3.3837
RMSE (Ave of axes) (BPM)	4.5766	4.4396	3.6889	4.2350 ± 0.4779

IV. DISCUSSION

This paper presents a comparison of the four IMU mounting locations of the head, shown in Fig. 1, for the purpose of BCG-based HR detection. We performed the KF-based HR estimation algorithm on two sets of IMU signals (from the accelerometer and gyroscope) for a comprehensive comparison.

Based on the results presented in Table 1, the neck location of the head outperformed other locations with RMSE of 8.3980 and 4.2350 with respect to the ECG reference and the average

of IMU signals, respectively. Note that these results are derived from accelerometer signals, highlighting the significance of the accelerometer over the gyroscope for this particular location. However, at the other three locations, the RMSE values from gyroscope signals showed better results, indicating that the gyroscope is more crucial than the accelerometer in detecting HR in non-neck locations. This discrepancy could be attributed to the different blood flow dynamics in the arteries in the head and neck. Therefore, a specific set of signals may better detect the micromovements caused by the cardiovascular activity in different locations when using BCG. Furthermore, as shown in this table, the estimation error is lower at the temple and the front of the ear on the zygomatic bone compared to the back of the ear on the mastoid. Thus, the most effective head mounting locations for detecting HR, in descending order of accuracy, are the neck on the carotid artery, the temple, the front of the ear on the zygomatic bone, and the back of the ear on the mastoid.

Table 2 demonstrates the RMSE of individual accelerometer signals at the neck location. Among these, the z-axis shows the least error, outperforming the other axes. Both the y-axis and the “n”-axis also yields acceptable results in two of the trials. However, the x-axis, with a mean RMSE of 17.9010, is not as accurate. According to the coordinates presented in Fig. 1, the x-axis of the accelerometer is neither perpendicular nor parallel to the neck, which compromises its ability to detect the micromovements of the artery via BCG. In contrast, the z-axis is perpendicular to the artery and thus shows the best results. This may be due to the IMU device being pressed against the carotid artery location along the z-axis, which could enhance the accuracy of measurements along this axis.

One of the main aims of this pilot study was to develop an accurate heart rate estimation method by measuring the small vibrations on the skin due to cardiac activities. However, despite employing various signal processing algorithms, the proposed method could be prone to errors because of motion artifacts such as head movements, breathing, swallowing, and blinking which could potentially affect heart rate detection. This issue is well-documented in literature, where reported RMSEs from CPAP masks in the lying position were 4.29 BPM and 3.28 BPM using custom weighted average algorithm and Kalman filter algorithm, respectively [17]. Hence, the RMSE of 4.23 BPM in the seated position in our study is consistent with similar studies regarding the body movements.

The main limitation of our pilot study was the small dataset size, which could impact the performance and generalizability of the proposed method. One concern, aside from variations in gender, weight, height, and age among participants, is that the position of the heart (heart’s angle) may vary among individuals. This could lead to different optimal axes in some participants, even though the overall algorithm should accurately and robustly estimate heart rate by utilizing four signals. Therefore, future research will aim to expand the dataset to a more diverse and larger sample size as well as including different activities such as standing and walking to validate the developed algorithm under various daily conditions. Furthermore, we plan to develop an automatic axis selection algorithm to exclude unnecessary axes from the dataset, thereby improving the HR estimation accuracies via IMU-based BCG.

V. CONCLUSION

This paper introduced a heart rate estimation method using Kalman filter data fusion technique applied to signals from accelerometers and gyroscopes mounted at four distinct head locations for IMU-based BCG. Our results show that accelerometer signals, particularly those from the z-axis that correspond to the perpendicular direction to the carotid artery, provide the most accurate heart rate detection when the sensor is located on the neck. Conversely, gyroscope signals yield superior heart rate estimations at the other three examined locations: the temple, the front of the ear on the zygomatic bone, and the back of the ear on the mastoid. These results underscore the significance of sensor placement on the head in optimizing heart rate monitoring accuracy using IMU-based BCG in the design of future head-mounted wearable devices. The result of this study could eventually be used in designing the form factor of the future earbuds and headphones products to continuously measure HR and HRV.

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